Volatility Surface Modelling and Forecasting

From Lévy Processes to Deep Learning

Amos Anderson

Stony Brook University
Applied Mathematics and Statistics

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The Data: GLD Options on 18 July 2025

A real option chain with 11 maturities and 877 quotes

Why GLD?

- Liquid gold ETF, tight bid-ask spreads
- 11 maturities: 8 DTE to 127 DTE
- $\sim$80 strikes per maturity (call + put)
- Source: `Barchart.com` end-of-day quotes

Key parameters:

- Spot: $S_0 = \$308.39$
- Moneyness range: $K/S_0 \in [0.924, 1.064]$
- Risk-free rate: interpolated Treasury curve (4.27% – 4.46%)
- Dividend yield: $q = 0.4\%$ (GLD expense ratio)

What We Observe

The market does not trade at a single implied volatility. Instead, every strike–maturity pair carries its own $\sigma_{\mathrm{imp}}(K, T)$. Our job is to model this **surface**.

877 contracts
11 expirations
1 common benchmark

The Black-Scholes Benchmark

How wrong is a single flat volatility?

What Black-Scholes assumes:

- One constant σ for **all** strikes and maturities
- Log-normal returns: no skew, no smile, no term structure
- Complete market: unique risk-neutral measure $\mathbb{Q}$

Best single volatility (least-squares):

$$\hat{\sigma}_{\text{flat}} = \arg\min_{\sigma} \sum_{i=1}^{877} \left(C_{\text{BS}}^{(i)}(\sigma) - C_{\text{mid}}^{(i)} \right)^2 = \boxed{16.20\%} \tag{1}$$

Overall pricing error: $\text{MAPE} = 8.32\%$

The Benchmark Logic

We do not advocate BS. We use it as a **deliberately naive floor**. Any model that cannot beat 8.32% MAPE is not economically viable.

Where Flat Volatility Fails

Four stylised facts that every model must explain

Stylised Fact	What BS Predicts	What Market Does	MAPE
1. **Skew** (OTM put premium)	σ_{imp} flat in K	Put $K/S = 0.944$: $+0.51$ pp vs ATM	8.60%
2. **Term structure**	σ_{imp} constant in T	Short ATM 13.65% $\rightarrow$ Long ATM 16.12%	8.02%
3. **Short-dated steepness**	Skew $\propto 1/\sqrt{T}$	Short skew is **5.1**$\times$ steeper than long	10.85%
4. **Smile asymmetry**	Gaussian symmetry	Call $K/S = 1.057$: $+1.30$ pp vs ATM	7.60%

The flat-vol assumption misses every directional feature of the market.

Visual Evidence: The BS Failure Dashboard

IV deviation from the best flat vol ($\hat{\sigma} = 16.20\%$)

output/05_flatvol_deviation.png

The Implied Volatility Surface

The empirical object: $\sigma_{\mathrm{imp}}(K, T)$ for GLD calls

output/01_surface_3d.png

Term Structure and Skew Slices

Two cross-sections of the same surface

Term Structure

output/04_term_structure.png

Skew by Maturity

output/03_skew_smile.png

What the Failure Tells Us About Model Design

Each stylised fact points to a different modelling strategy

Four stylised facts that flat volatility misses

1. Skew
$\rho < 0$ correlation

2. Term structure
Mean-reverting vol

3. Short steepness
Jumps needed

4. Smile asymmetry
Heavy tails

Generation 1
Heston SV
CIR + correlation

Generation 2
Lévy (NIG, VG)
Heavy-tailed returns

Generation 3
Deep Learning
ConvLSTM + physics

Variance swaps complete the market

Escher transform selects $\mathbb{Q}$

Learns directly from data

The benchmark surface is set. Next: can Heston (Gen 1) clear the 8.32% MAPE hurdle?

Generation 1: Heston Stochastic Volatility

Making volatility itself a random process

The Heston dynamics:

$$\mathrm{d}S_t = (r-q)S_t\,\mathrm{d}t + \sqrt{v_t}\,S_t\,\mathrm{d}W_t^1$$
$$\mathrm{d}v_t = \kappa(\theta - v_t)\,\mathrm{d}t + \sigma_v\sqrt{v_t}\,\mathrm{d}W_t^2$$

with $\mathrm{d}\langle W^1, W^2\rangle = \rho\,\mathrm{d}t$.

The economic mechanism:

- $\rho < 0$: spot drops $\Rightarrow$ variance rises $\Rightarrow$ OTM puts expensive
- CIR: variance mean-reverts to θ at speed κ
- Closed-form characteristic function $\Rightarrow$ FFT pricing

Fitted Parameters (GLD)

Parameter	Value
v_0 (initial var)	0.0220
κ (speed)	10.00
θ (long var)	0.0308
σ_v (vol of vol)	0.6598
ρ (correlation)	+0.24

Feller: $2\kappa\theta - \sigma_v^2 = 0.18 > 0$ ✓

Heston Results Table

29.9% better than BS overall, but deep OTM puts worsen

Moneyness Bucket	BS Flat Vol	Heston SV	Winner
Deep OTM Put ($K/S \leq 0.95$)	8.36%	12.19%	BS
OTM Put ($0.95 < K/S < 0.98$)	11.14%	6.08%	Heston
ATM ($0.98 \leq K/S \leq 1.02$)	8.01%	4.97%	Heston
OTM Call ($1.02 < K/S < 1.05$)	6.12%	5.43%	Heston
Deep OTM Call ($K/S \geq 1.05$)	7.29%	3.57%	Heston
Overall MAPE	**8.32%**	**5.83%**	**Heston**

- **The problem:** CIR diffusion cannot generate enough short-maturity skew for deep OTM puts. The variance process needs time to react—but short-dated options expire first.
- **The solution:** Replace the Gaussian driver with jumps (Lévy processes, Generation 2).

Visual Comparison: BS vs Heston Implied Vol

Heston captures the bulk of the smile, but not the extreme short end

BS Flat Vol (16.20%)

output/01_surface_3d.png

Heston SV (fitted)

output/heston_surface_3d.png

Visual Comparison: BS vs Heston Implied Vol

Contour Plots

BS Flat Vol (16.20%)

output/02_surface_2d_contour.png

Heston SV (fitted)

output/heston_surface_2d_contour.png

Generation 1: The SVJ Extension

Adding compound Poisson jumps to Heston (Bates 1996)

The SVJ dynamics:

$$\mathrm{d}S_t = (r - q - \lambda\mu_J)\, S_t\, \mathrm{d}t + \sqrt{v_t}\, S_t\, \mathrm{d}W_t^1 + S_t\,(e^J - 1)\, \mathrm{d}N_t$$

$$\mathrm{d}v_t = \kappa(\theta - v_t)\, \mathrm{d}t + \sigma_v \sqrt{v_t}\, \mathrm{d}W_t^2$$

$$J \sim \mathcal{N}(\mu_J, \sigma_J^2), \qquad \mathrm{d}N_t \sim \mathrm{Poisson}(\lambda\, \mathrm{d}t)$$

Calibrated jump parameters (Heston params frozen): $\lambda = 0.0102$, $\mu_J \approx 0$, $\sigma_J = 0.01$

Metric	**BS Flat Vol**	**Heston SV**	**Heston + SVJ**
Overall MAPE	8.32%	5.83%	**5.83%**
Deep OTM put MAPE	8.36%	12.19%	**12.19%**

The jump component did not activate.

- With GLD's narrow strike range ($K/S \in [0.924, 1.064]$), the optimizer finds $\lambda \approx 0$, $\mu_J \approx 0$: *no jumps needed to fit the observed prices.*
- This confirms Gatheral (2006): finite-activity Poisson jumps cannot reproduce the continuous short-dated skew. **Infinite-activity is needed**—Lévy processes.

Generation 2: Lévy Processes

Infinite-activity jumps to capture the short-dated skew

The problem Generation 1 left unsolved:

- Heston diffusion needs *time* to generate variance — short-dated skew too shallow
- SVJ adds *finite* Poisson jumps (λ too low) — still not enough
- Empirical skew scales as T^H, $H < 1/2$; diffusion gives $T^{1/2}$

The Lévy insight: Replace the Gaussian underlying process with a process that has **infinite activity**—jumps at *every* time scale, not just discrete arrivals.

Lévy-Khinchine triplet (γ, σ, ν)**:**

$$\varphi(u) = \exp\left\{ t \left[i\gamma u - \frac{\sigma^2 u^2}{2} + \int_{\mathbb{R}} (e^{iux} - 1 - iux 1_{|x|<1})\, \nu(\mathrm{d}x) \right] \right\} \tag{2}$$

Market incompleteness: Jumps cannot be hedged by trading S_t alone. The Esscher transform selects a unique risk-neutral measure $\mathbb{Q}_\theta$ via exponential tilting.

Three Lévy Specifications

NIG, VG, and BGM — different tail behaviours, same FFT pipeline

	NIG	VG	BGM
Origin	Barndorff-Nielsen (1997)	Madan-Carr-Chang (1998)	Michaelides-Kirkby (2024)
Time change	Inverse Gaussian	Gamma	Two independent Gammas
Parameters	$\alpha, \beta, \delta, \mu$	σ, ν, θ	$\alpha_\pm, \beta_\pm$
Tail control	Symmetric via β	Symmetric via ν	**Independent** $\alpha_\pm, \beta_\pm$
CF complexity	Bessel K_1	Elementary	Elementary (product)
Pricing	FFT or Bessel explicit	FFT	FFT

The BGM advantage: Independent left/right tail parameters mean the model can fit gold's two-sided crash risk without forcing symmetry.

1. Normal Inverse Gaussian (NIG)

Barndorff-Nielsen (1997) — Brownian motion subordinated by Inverse Gaussian time

The intuition. Start with standard Brownian motion with drift θ:

$$X_t \;=\; \theta t + \sigma W_t$$

Now replace calendar time t with a random clock T_t that follows *Inverse Gaussian* distribution (also called the Wald distribution). The resulting process is **NIG**:

$$L_t \;=\; \theta\, T_t + \sigma\, W_{T_t}$$

Why this matters for options. The Inverse Gaussian clock has **heavier tails** than the normal clock. When time itself is random and skewed, the returns inherit skewness and kurtosis even though the underlying Brownian motion is Gaussian.

Characteristic function (closed form):

$$\varphi_{\mathrm{NIG}}(u) = \exp\left\{ i\mu u t + \delta t\left(\sqrt{\alpha^2 - \beta^2} - \sqrt{\alpha^2 - (\beta + iu)^2} \right) \right\}$$

Parameters:

$\alpha > 0$	Tail heaviness (steepness of the smile)
$\lvert\beta\rvert < \alpha$	Skewness ($\beta < 0$ for negative skew)
$\delta > 0$	Scale (overall volatility level)
$\mu \in \mathbb{R}$	Location (drift adjustment)

Key property: NIG is a *normal variance-mean mixture*. The density involves the modified Bessel function K_1, but the characteristic function above is elementary enough for FFT pricing.

2. Variance Gamma (VG)

Madan, Carr & Chang (1998) — Brownian motion subordinated by Gamma time

The intuition. Same recipe as NIG: start with Brownian motion, but now replace the clock with a *Gamma* process G_t instead of Inverse Gaussian. The Gamma clock is **pure-jump with infinite activity** — it ticks at every instant, but most ticks are infinitesimally small.

$$L_t \;=\; \theta\, G_t + \sigma\, W_{G_t}, \qquad G_t \sim \Gamma\!\left(\frac{t}{\nu},\, \nu\right)$$

Why this matters for options. The Gamma subordinator produces **finite moments but infinite activity**. The model generates a continuous stream of small jumps that creates the steep short-dated skew — exactly what Heston diffusion misses.

Characteristic function (elementary):

$$\varphi_{\mathrm{VG}}(u) \;=\; \left(1 - i\theta\nu u + \frac{\sigma^2\nu u^2}{2}\right)^{-t/\nu}$$

Parameters:

$\sigma > 0$	Volatility of the Brownian motion
$\nu > 0$	Variance rate of the Gamma clock
	(as $\nu \to 0$, VG $\to$ Black-Scholes)
$\theta \in \mathbb{R}$	Drift of the subordinated Brownian motion
	($\theta < 0$ produces negative skew)

Key property: VG is the *simplest* exponential Lévy model for finance — no Bessel functions, no hypergeometrics. The CF is a power function, making calibration fast and FFT pricing trivial.

3. Bilateral Gamma Motion (BGM)

Michaelides & Kirkby (2024) — independent Gamma clocks for each tail

The intuition. NIG and VG use a *single* time change for the entire return distribution. BGM breaks the clock into two **independent** Gamma processes:

$$L_t = X_t^+ - X_t^-, \qquad X_t^\pm \sim \Gamma(\alpha_\pm t, \beta_\pm)$$

X^+ drives **positive** jumps (upside), X^- drives **negative** jumps (downside). They are completely decoupled.

Why this matters for gold (GLD). Equity indices typically exhibit *one-sided* skew — OTM puts are expensive, OTM calls are cheap. Gold is different: both supply shocks and safe-haven demand can generate large moves. A model with **independent tail parameters** can fit two-sided tail risk without forcing symmetry.

Characteristic function (product of two Gamma CFs):

$$\varphi_{\mathrm{BGM}}(u) = \left(1 - \frac{iu}{\beta_+}\right)^{-\alpha_+ t} \left(1 + \frac{iu}{\beta_-}\right)^{-\alpha_- t}$$

Parameters:

α_+, α_-	Shape (tail heaviness of upside / downside)
β_+, β_-	Rate (frequency of positive / negative jumps)
	Larger $\beta_+ \Rightarrow$ fewer large upward jumps
	Smaller $\beta_- \Rightarrow$ fatter left tail (crash risk)

Key property: BGM is the only model among the three where **skewness and kurtosis are separately controlled for each tail**. This is economically meaningful for commodities and cryptocurrencies, where two-sided tail risk is the rule, not the exception.

Generation 2: Expected vs. Actual Performance

Literature expectation (Carr–Madan 2002, Schoutens 2003). On equity-index options (SPX), infinite-activity Lévy processes typically *beat* stochastic-volatility models on the short-dated skew because semi-heavy tails generate a steeper OTM-put smile than a CIR variance process alone.

What we found on GLD (2025-07-18).

Model	IV-MAPE (%)	Price-MAPE (%)	RMSE ($)
Black–Scholes (flat vol)	8.32	8.32	—
Heston (stochastic vol)	**5.83**	**5.83**	—
NIG (infinite-activity)	8.22	17.66	0.29
VG (infinite-activity)	8.37	13.58	0.36
BGM (infinite-activity)	8.35	18.96	0.29

No improvement. All three Lévy models match (but do not beat) flat-vol BS in IV space, and are significantly *worse* than Heston.

Why Lévy underperforms on GLD: three explanations

1. **Boundary-seeking = Gaussian collapse.** Differential evolution pushes parameters to box edges:

$$\text{NIG: } \alpha = 24.7 \text{ (ub} = 25), \quad \text{VG: } \nu = 0.01 \text{ (lb)}, \quad \text{BGM: } \alpha_- = 20 \text{ (ub)}.$$

   As $\alpha \to \infty$ (NIG) or $\nu \to 0$ (VG), the process collapses to Brownian motion. The data is telling the optimizer: "I do not need jumps."

2. **Narrow moneyness range.** GLD strikes span only $K/S \in [0.924, 1.064]$. There are no deep OTM wings ($K/S < 0.8$ or > 1.2) where semi-heavy tails would generate the characteristic steep skew that Lévy models are famous for.

3. **IV-space vs. Price-space pathology.** In IV space the Lévy models match BS ($\sim$8%). In price space they degrade to 13–19% because deep OTM puts have prices $\ll$ \$1; tiny dollar errors become enormous relative errors when the denominator is small.

Visual check 1/3: Model IV vs. Market IV

output/gen2_levy_iv_fit.png

Visual check 2/3: Model Price vs. Market Price

output/gen2_levy_scatter_fit.png

Visual check 3/3: Error Distributions

output/gen2_levy_error_distributions.png

What we have established so far.

- BS flat-vol fails (8.32% MAPE) — the smile exists.
- Heston captures it (5.83% MAPE) — stochastic volatility is the *dominant* mechanism for this surface.
- Lévy adds no marginal value (8.2–8.4% IV-MAPE) — the data does not exhibit the extreme kurtosis or wide moneyness wings that infinite-activity jumps were designed for.

The unanswered question. If parametric models (BS, Heston, NIG, VG, BGM) are all *too rigid* or *too expressive* for this surface, can we learn the mapping $(K, T) \mapsto \sigma_{\text{impl}}$ directly from the data without imposing a functional form?

Generation 3: Deep Learning.

- **ConvLSTM** — spatiotemporal evolution of the surface.
- **Attention-LSTM** — cross-dependencies across strikes and maturities.
- **Physics-informed Transformer** — hard-wires no-arbitrage constraints (calendar spread, butterfly) into the architecture.